

PROFESSIONAL SUMMARY

Software Engineer (M.S. Information Technology, Carnegie Mellon University) with 6+ years of experience architecting and optimizing secure, high-availability backend systems. Improved performance of a global financial data verification system by **~95%** by redesigning processing logic and applying parallelized computation, enabling reliable execution on large-scale datasets. Strong in **Java, Spring Boot, and SQL**, with hands-on experience designing APIs, implementing authentication (2FA/RBAC), and ensuring data integrity. Currently **expanding into full-stack architecture with Next.js and React** to complement a deep background in scalable backend systems.

- Java Backend Development
- RESTful API Development
- SQL Database Management
- Linux System Administration
- Data Validation/Management
- Authentication/Access Control
- Cross-Functional Coordination
- System Performance Optimization
- Software Testing

EDUCATION | CERTIFICATIONS

Master of Science (M.S.) in Information Technology | Carnegie Mellon University | 2019
Bachelor of Science (B.S.) in Information Technology | University of Rwanda | 2016

TECHNICAL COMPETENCIES

Backend & APIs:

Java, Spring Boot, Spring MVC, Spring Data JPA, REST APIs

Data & Databases:

SQL (PostgreSQL, Oracle, MS SQL Server), Data Validation, Data Integrity

Systems & Performance:

Performance Optimization, Scalable System Design, Parallel Processing

Security & Authentication:

Authentication, Authorization, 2FA

Tools & Infrastructure:

Docker, Git, CI/CD (GitHub Actions), Linux

Programming Languages:

Java, Python, JavaScript, React, Typescript, C/C++, C#, HTML/CSS

PROFESSIONAL EXPERIENCE

World Bank Group | Washington, DC

2019 – 2023

Backend Developer

Designed backend services, APIs, and authentication controls while coordinating across teams to ensure secure access, system reliability, and performance optimization within enterprise environments.

- **Security & Authentication:** Engineered RESTful APIs and strengthened system integrity by implementing 2FA and Role-Based Access Control (RBAC), ensuring compliance with international security standards.
- **Logic Redesign & Optimization:** Optimized a global data verification system by redesigning processing logic, partitioning datasets, and parallelizing workloads, achieving **~95%** performance improvement and enabling reliable execution on large-scale data.
- **Cross-Functional Alignment:** Led technical discussions between engineering, QA, and business units to standardize system access and ensure seamless deployment of performance upgrades.
- **Operational Integrity:** Supported large-scale analytical workflows by ensuring data persistence and reliability within complex, multi-country database environments.

Pitech Rwanda | Remote
Backend Software Engineer

2023 – 2025

Developed and supported backend systems for enterprise property valuation platforms, focusing on operational continuity and data accuracy.

- **Backend Maintenance:** Enhanced core system functionality using **Java and SQL**, ensuring daily reporting workflows remained reliable for high-stakes property data.
- **System Reliability:** Diagnosed and resolved production bottlenecks by collaborating across engineering and QA teams, implementing fixes to restore service without disrupting business operations.
- **Technical Guidance:** Provided workflow training and technical documentation to internal teams, improving system adoption and ensuring consistent execution of backend processes.
- **Regional Product Design:** Contributed to design initiatives for new digital products, coordinating with stakeholders across time zones to align technical solutions with regional requirements.

Carnegie Mellon University | Pittsburgh, PA
Technical Research and Teaching Assistant

2019

Assisted instruction, supported students, maintained academic records, and ensured research documentation accuracy while strengthening computing education and cybersecurity research support initiatives.

- **Instructional Support:** Assisted in teaching computer science fundamentals, including algorithms and debugging, reinforcing technical logic for a diverse student body.
- **Information Management:** Translated and summarized cybersecurity research data, ensuring the accuracy and organization of digital records for sensitive research initiatives

PERSONAL SOFTWARE PROJECTS

- **Recipe Book Application (In Progress):** Developing a frontend application using React and Tailwind CSS with planned backend integration and database implementation to support full-stack functionality and data persistence.
- **Service Marketplace Platform (In Progress):** Building a full-stack marketplace application with Next.js and TypeScript frontend and Spring Boot backend, implementing authentication workflows and service listing functionality.
- **Technology Stack Portfolio Web App:** Created a static web application using HTML, CSS, and JavaScript to showcase technical stack preferences and demonstrate frontend development capabilities.